SafeCart: An AI-Powered Allergen Detection System

DA627 - Multimodal Generative AI Course Project | Mid-Evaluation Progress Report

Rishita Agarwal (220150016) | IIT Guwahati

Project Overview

SafeCart

An AI-powered system that automatically identifies potential allergens from food product packaging images using multimodal AI techniques combining Optical Character Recognition and Natural Language Processing.

Core Innovation

Real-time visual analysis of product packaging with personalized allergen filtering based on user profiles, providing instant safety classification.

The Challenge

200M+ People Affected

Food allergies represent a significant public health challenge affecting over 200 million people globally

10.8% U.S. Adults

Over 10.8% of adults in the United States suffer from at least one food allergy

30-60 Minutes Per Trip

Allergy-sensitive individuals spend an average of 30-60 minutes per grocery shopping trip reading product labels

38% Severe Reactions

Approximately 38% of food allergy sufferers have experienced a severe allergic reaction

The complexity of food labels, hidden allergens under technical ingredient names, and inconsistent labeling practices create significant barriers to safe food selection.

Project Objectives

01

Real-Time Pipeline

Develop a real-time pipeline combining OCR and NLP for allergen detection

02

Comprehensive Database

Build a comprehensive allergen keyword database with synonym mapping

03

Personalized Filtering

Implement personalized allergen filtering based on user profiles

04

Web Interface

Create a web-based interface with intuitive safety indicators

05

Performance Evaluation

Evaluate system performance on real product images

System Architecture

Image Processing

OpenCV preprocessing: grayscale conversion, noise reduction, contrast enhancement

OCR Engine

EasyOCR text extraction from product images with multiple font support

NLP Processing

Pattern matching against allergen database with synonym mappings

Classification

Confidence scoring and safety classification with personalized filtering

Technology Stack

Backend Framework

FastAPI (Python) for web application

OCR Engine

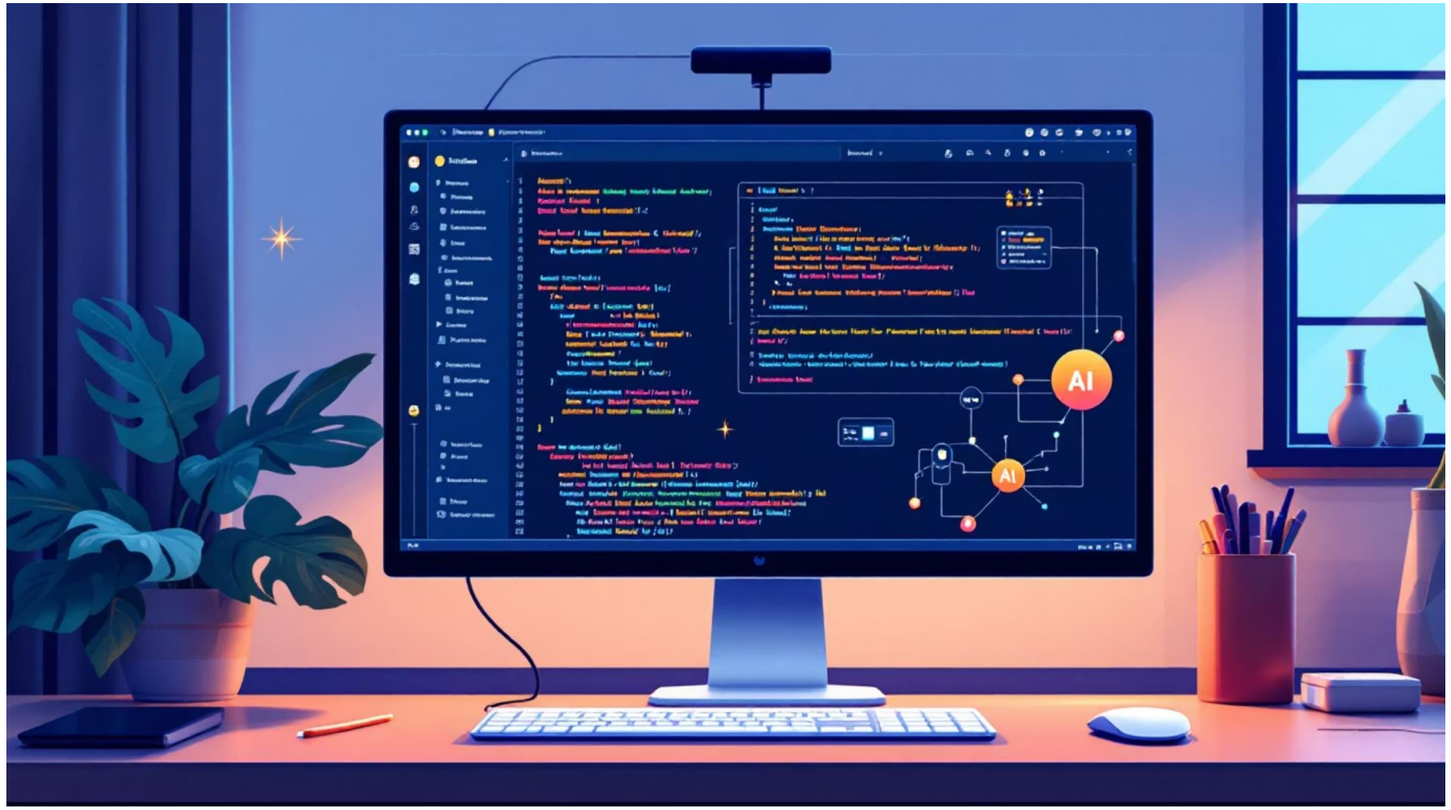
EasyOCR for text extraction

Image Processing

OpenCV for preprocessing

Frontend

HTML, CSS, JavaScript for interface



Implementation Progress

1

OCR Module

- Image preprocessing pipeline with OpenCV
- Text extraction from product images
- Handling of various image qualities

2

Allergen Database

- 8 major allergen categories
- Synonym mappings for technical names
- Common variations and spellings

3

NLP Module

- Text tokenization and normalization
- Keyword matching against database
- Confidence score calculation

4

Web Interface

- Image upload functionality
- Allergen input for personalization
- Results display with safety indicators

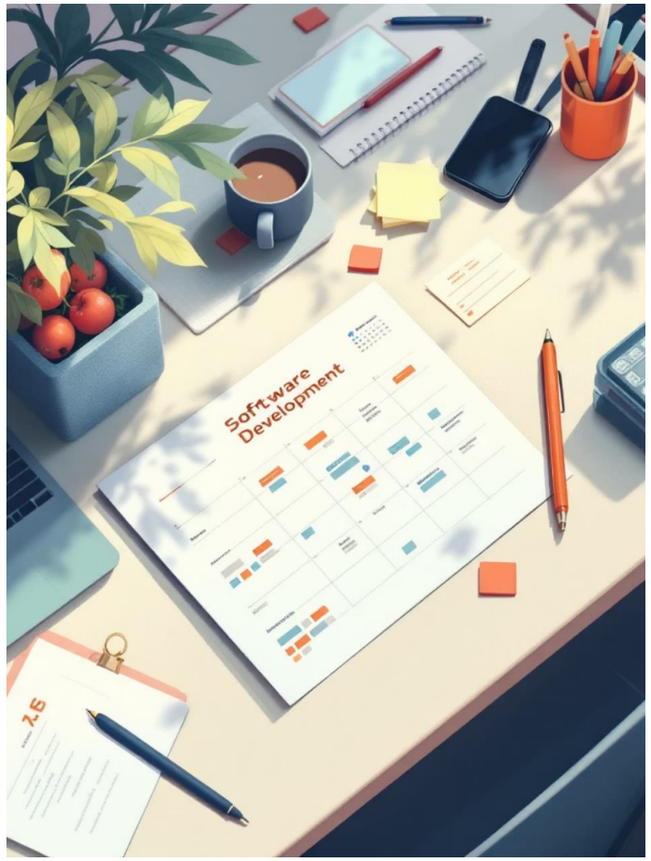
Remaining Work

Immediate Tasks

- Expand test dataset with more product images
- Improve OCR preprocessing for challenging images
- Add more allergen synonyms to database
- Implement grocery list generation module

Evaluation Tasks

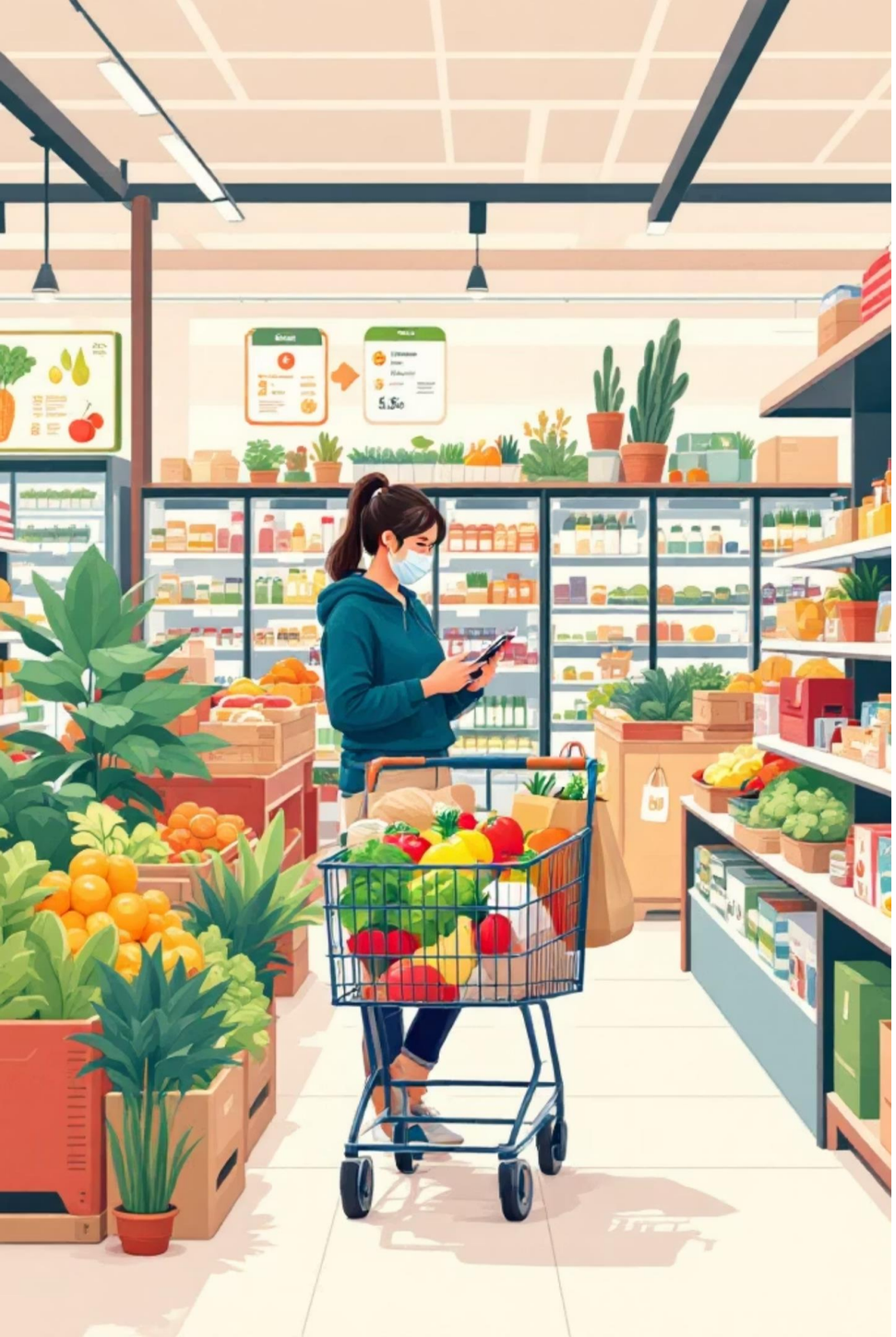
- Conduct systematic evaluation on full test dataset
- Calculate precision, recall, and F1-score metrics
- Measure processing time performance
- Analyze failure cases and limitations



Timeline

Week 1-2: Dataset expansion and OCR improvements

Week 2: Experimental evaluation. Final report and presentation preparation



Conclusion

On Track

Core pipeline combining OCR and NLP has been implemented and demonstrates functional allergen detection capability

Next Steps

Systematic evaluation, performance optimization, and comprehensive documentation for final submission

Impact

Addressing real-world challenge of safe grocery shopping for allergy-sensitive individuals with personalized AI assistance

Thank you!